SQL CHEAT SHEET

100% INTERVIEW READY

THE 20 SQL PATTERNS YOU NEED TO MASTER

> Don’t memorize queries. Master the exact patterns used by Goldman Sachs, Amazon & Google.

START LEARNING →

Anandnarayanan S

Problem 1/20

WEIGHTED AVG

GOLDMAN SACHS

Weighted Score

Calculate the weighted average score for each department.

SAMPLE INPUT

Dept	Score	Wgt
HR	80	0.3
HR	90	0.7
IT	75	0.5

EXPECTED OUTPUT

Dept	W_Avg
HR	87.0
IT	80.0

SOLUTION

MYSQL

```mysql
-- Sum Product of Score * Weight / Sum Weights
SELECT
    Department,
    SUM(Score * Weight) / SUM(Weight) AS W_Avg
FROM Scores
GROUP BY Department;
```

Next: Duplicates →

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Problem 2/20

GROUP BY

GOOGLE

Find Duplicates

Identify emails that appear more than once in the user table.

SAMPLE INPUT

ID	Email
1	a@x.com
2	a@x.com
3	b@x.com

EXPECTED OUTPUT

Email	Cnt
a@x.com	2

SOLUTION

SQL SERVER

```sql
-- Group and Filter Count > 1
SELECT Email, COUNT(*) as Cnt
FROM Users
GROUP BY Email
HAVING COUNT(*) > 1;
```

Next: Mode →

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Problem 3/20

MODE

DOORDASH

Most Frequent

Find the mode (most frequent value) of a response column.

SAMPLE INPUT

ID	Response
1	Good
2	Bad
3	Good

EXPECTED OUTPUT

Response	Freq
Good	2

SOLUTION

MYSQL

```mysql
-- Order by frequency count and limit 1
SELECT
    Response,
    COUNT(*) AS Freq
FROM Survey
GROUP BY Response
ORDER BY Freq DESC
LIMIT 1;
```

Next: Pivot Data →

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Problem 4/20

PIVOT / CASE

AMAZON

Pivot Data

Pivot monthly revenue rows into columns for each month.

SAMPLE INPUT

Mth	Rev
Jan	100
Feb	200

EXPECTED OUTPUT

Jan_Rev	Feb_Rev
100	200

SOLUTION

MYSQL

```mysql
-- Conditional Aggregation to Pivot
SELECT
  SUM(CASE WHEN Mth='Jan' THEN Rev END) AS Jan_Rev,
  SUM(CASE WHEN Mth='Feb' THEN Rev END) AS Feb_Rev
FROM Revenue;
```

Next: Time Buckets →

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Problem 5/20

TIME BUCKET

UBER

15 Min Intervals

Group sensor data by 15-minute intervals and calculate average.

SAMPLE INPUT

Time	Val
10:05	50
10:10	60
10:20	55

EXPECTED OUTPUT

Interval	AvgVal
10:00	55.0
10:15	55.0

SOLUTION

SQL SERVER

```sql
-- Floor time to nearest 15 mins
SELECT
    DATEADD(MINUTE,
      DATEDIFF(MINUTE, 0, Timestamp) / 15 * 15, 0
    ) AS Interval,
    AVG(Value) as AvgVal
FROM Sensors
GROUP BY 1;
```

Next: Extract Domain →

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Problem 6/20

SUBSTRING

SNOWFLAKE

Extract Domain

Extract the domain name from an email address column.

SAMPLE INPUT

ID	Email
1	dev@meta.com
2	ceo@ibm.com

EXPECTED OUTPUT

ID	Domain
1	meta.com
2	ibm.com

SOLUTION

POSTGRESQL

```sql
-- Substring from @ index to end
SELECT
  Email,
  SUBSTRING(Email, CHARINDEX('@', Email) + 1, LEN(Email))
  AS Domain
FROM Users;
```

Next: Filtered Sums →

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Problem 7/20

HAVING

WALMART

Filtered Sums

Aggregate sales by Region/Product but only include sales > 1500.

SAMPLE INPUT

Reg	Prod	Amt
N	Lap	2000
N	Ph	1000

EXPECTED OUTPUT

Reg	Prod	Total
N	Lap	2000

SOLUTION

MYSQL

```mysql
-- Filter BEFORE grouping
SELECT
    Region, Product,
    SUM(SalesAmount) AS Total
FROM Sales
WHERE SalesAmount > 1500
GROUP BY Region, Product;
```

Next: Running Total →

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Problem 8/20

SUM OVER

STRIPE

Running Total

Calculate a cumulative running total of revenue by date.

SAMPLE INPUT

Date	Rev
Jan 1	10
Jan 2	20
Jan 3	30

EXPECTED OUTPUT

Date	Cumul
Jan 1	10
Jan 2	30
Jan 3	60

SOLUTION

SQL SERVER

```sql
-- Unbounded Preceding Sum
SELECT Date, Rev,
  SUM(Rev) OVER (ORDER BY Date) as Cumul
FROM Sales;
```

Next: Unique Pairs →

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Problem 9/20

CONCAT

SHOPIFY

Unique Pairs

Count distinct combinations of Customer and Product.

SAMPLE INPUT

C_ID	P_ID
1	A
1	A
1	B

EXPECTED OUTPUT

UniquePairs
2

SOLUTION

MYSQL

```mysql
-- Concat columns to create unique key
SELECT
    COUNT(DISTINCT
      CONCAT(CustomerID, '-', ProductID)
    ) AS UniquePairs
FROM Orders;
```

Next: 2nd Highest →

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Problem 10/20

DENSE_RANK

MICROSOFT

2nd Highest

Find the employee with the second highest salary.

SAMPLE INPUT

Emp	Sal
A	100
B	200
C	300

EXPECTED OUTPUT

Emp	Sal
B	200

SOLUTION

SQL SERVER

```sql
-- Rank and select where Rank = 2
WITH Ranked AS (
  SELECT *,
  DENSE_RANK() OVER(ORDER BY Sal DESC) as rnk
  FROM Emp
)
SELECT Emp, Sal FROM Ranked WHERE rnk = 2;
```

Next: Case Buckets →

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Problem 11/20

CASE GROUP

AMEX

Custom Buckets

Group transactions into Low, Medium, High based on amount.

SAMPLE INPUT

ID	Amt
1	50
2	500

EXPECTED OUTPUT

Cat	Cnt
Low	1
High	1

SOLUTION

MYSQL

```sql
-- Group by Derived Column
SELECT
  CASE
    WHEN Amount < 100 THEN 'Low'
    ELSE 'High'
  END AS Cat,
  COUNT(*) as Cnt
FROM Txns
GROUP BY Cat;
```

Next: YoY Growth →

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Problem 12/20

LAG

META

YoY Growth

Calculate Year-Over-Year percentage growth in revenue.

SAMPLE INPUT

Yr	Rev
2023	100
2024	120

EXPECTED OUTPUT

Yr	Growth
2023	NULL
2024	20%

SOLUTION

POSTGRESQL

```postgresql
-- (Curr - Prev) / Prev
WITH Prev AS (
  SELECT Yr, Rev,
  LAG(Rev) OVER(ORDER BY Yr) as LastYr
  FROM Sales
)
SELECT Yr, (Rev-LastYr)*100.0/LastYr as Growth
FROM Prev;
```

Next: Median →

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Problem 13/20

PERCENTILE

LINKEDIN

Find Median

Calculate the median score (50th percentile) of students.

SAMPLE INPUT

ID	Score
1	70
2	80
3	90

EXPECTED OUTPUT

Median
80

SOLUTION

SQL SERVER

```sql
-- Select middle row using ranking
WITH Ranked AS (
  SELECT Score,
  ROW_NUMBER() OVER(ORDER BY Score) rn,
  COUNT(*) OVER() cnt
  FROM Scores
)
SELECT AVG(Score) as Median FROM Ranked
WHERE rn IN ((cnt+1)/2, (cnt+2)/2);
```

Next: Top Per Group →

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Problem 14/20

ROW_NUMBER

UBER

Top Per Group

Find the most recent order for each user.

SAMPLE INPUT

User	Date
A	Jan 1
A	Jan 5
B	Feb 1

EXPECTED OUTPUT

User	Latest
A	Jan 5
B	Feb 1

SOLUTION

MYSQL

```mysql
-- Partition by User, Order by Date Desc
SELECT User, Date as Latest FROM (
  SELECT *,
  ROW_NUMBER() OVER(PARTITION BY User ORDER BY Date DESC)
rn
  FROM Orders
) t WHERE rn = 1;
```

Next: Subtotals →

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Problem 15/20

ROLLUP

AMAZON

Subtotals

Calculate subtotals and grand totals dynamically for Region sales.

SAMPLE INPUT

Reg	Prod	Amt
N	A	10

EXPECTED OUTPUT

Reg	Prod	Sum
N	A	10
N	NULL	10
NULL	NULL	10

SOLUTION

MYSQL

```mysql
-- Generate hierarchy totals
SELECT
    Region, Product,
    SUM(SalesAmount) as Sum
FROM Sales
GROUP BY ROLLUP(Region, Product);
```

Next: Data Gaps →

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Problem 16/20

LEAD

AIRBNB

Data Gaps

Identify days where no orders were placed (Gaps in sequence).

SAMPLE INPUT

OrderDate
Jan 1
Jan 2
Jan 5

EXPECTED OUTPUT

GapStart	GapEnd
Jan 2	Jan 5

SOLUTION

SQL SERVER

```sql
-- Check if Next Day > Current Day + 1
SELECT Date as GapStart, NextDate as GapEnd
FROM (
  SELECT Date,
  LEAD(Date) OVER(ORDER BY Date) as NextDate
  FROM Orders
) t
WHERE DATEDIFF(day, Date, NextDate) > 1;
```

Next: % Share →

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Problem 17/20

WIN RATIO

SALESFORCE

% Share

Calculate the percentage share of each product within its region.

SAMPLE INPUT

Region	Product	Amount
North	Laptop	20
North	Phone	80

EXPECTED OUTPUT

Region	Product	Pct_Share
North	Laptop	20.0
North	Phone	80.0

SOLUTION

SQL SERVER

```sql
-- Ratio to Report Logic
SELECT Region, Product, Amount,
  Amount * 100.0 /
  SUM(Amount) OVER(PARTITION BY Region)
  AS Pct_Share
FROM Sales;
```

Next: Coalesce →

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The 20 Pattern SQL Cheatsheet

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REPOST

Problem 18/20

COALESCE

NETFLIX

Handle Nulls

Return the first non-null contact method (Phone, then Email).

SAMPLE INPUT

ID	Ph	Em
1	NULL	a@x.com
2	123	b@x.com

EXPECTED OUTPUT

ID	Contact
1	a@x.com
2	123

SOLUTION

POSTGRESQL

```sql
-- First non-null value in list
SELECT
  ID,
  COALESCE(Phone, Email, 'No Contact') as Contact
FROM Users;
```

Next: Gini Coeff. →

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Problem 19/20

CUMULATIVE

WORLD BANK

Gini Coeff.

Calculate income inequality using the Lorenz Curve method.

SAMPLE INPUT

ID	Inc
1	5k
2	7k
3	2k

EXPECTED OUTPUT

Gini
0.24

SOLUTION

SQL SERVER

```sql
-- Complex Cumulative Stats
WITH Lorenz AS (
    SELECT Income,
    SUM(Income) OVER(ORDER BY Income) /
    SUM(Income) OVER() as CumShare
    FROM Incomes
)
SELECT 1 - 2 * SUM(CumShare) as Gini FROM Lorenz;
```

Next: Self Join →

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Problem 20/20

SELF JOIN

APPLE

Manager Sal.

Find employees who earn more than their managers.

SAMPLE INPUT

Emp	Sal	Mgr
A	100	B
B	90	NULL

EXPECTED OUTPUT

Emp
A

SOLUTION

MYSQL

```mysql
-- Join table to itself on Mgr ID
SELECT E.Name as Emp
FROM Emp E
JOIN Emp M ON E.MgrId = M.Id
WHERE E.Salary > M.Salary;
```

Next: MISSION ACCOMPLISHED →

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MISSION ACCOMPLISHED

You are Interview Ready!

20 PATTERNS MASTERED 100%

- [x] WEIGHTED AVG
- [x] GROUP BY
- [x] MODE
- [x] PIVOT / CASE
- [x] TIME BUCKET
- [x] SUBSTRING
- [x] HAVING
- [x] SUM OVER
- [x] CONCAT
- [x] DENSE_RANK
- [x] CASE GROUP
- [x] LAG
- [x] PERCENTILE
- [x] ROW_NUMBER
- [x] ROLLUP
- [x] LEAD
- [x] WIN RATIO
- [x] COALESCE
- [x] CUMULATIVE
- [x] SELF JOIN

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